perplexity

# Enhanced Feature Engineering for Power Demand Forecasting

Based on extensive research from academic papers and industry practices, here are comprehensive feature engineering suggestions to improve your power demand forecasting model:

### Core Weather Features

Your current model includes basic weather features, but research shows these advanced meteorological features significantly improve accuracy:

### Enhanced Weather Variables

- **Apparent Temperature/Heat Index**: Combines temperature and humidity to reflect perceived temperature[1][2]
- **Cooling/Heating Degree Days (CDD/HDD)**: Better captures temperature-demand relationship than raw temperature[3][4]
- **Wind Chill Factor**: Important for heating demand in cold climates[3]
- **Dew Point Temperature**: Critical for humidity-related energy demand[5]
- **Solar Radiation/Irradiance**: Affects cooling loads and solar generation[6][7]
- **Precipitation Intensity**: Heavy rain affects both demand patterns and renewable generation[8]
- **Atmospheric Pressure**: Influences weather patterns and comfort levels[5]

### Weather Interaction Features

- **Temperature × Humidity Interactions**: Crucial for air conditioning demand[9]
- **Temperature Variance**: Day-night temperature differences affect thermal inertia[10]
- **Weather Volatility**: Rolling standard deviation of weather parameters[8]

### Advanced Calendar and Temporal Features

### Holiday and Special Event Features

Research shows holiday effects can improve forecasting accuracy by up to 80%[11][12]:

- **Holiday Classification**: Fixed-date vs. weekday holidays[11]
- **Holiday Proximity**: Days before/after holidays (bridge effects)[13]

- **Holiday Type**: National, religious, local, school holidays[14]
- **Holiday Duration**: Single day vs. extended holiday periods[13]
- **Cultural Events**: Local festivals, sports events, elections[15]

### Enhanced Temporal Features

- **Daylight Savings Time**: Transition periods affect demand patterns[16]
- **Business/School Calendar**: Academic year patterns, vacation periods[17]
- **Peak/Off-Peak Indicators**: Time-of-use classifications[18]
- **Rush Hour Indicators**: Morning/evening peak periods[19]

### Economic and Socioeconomic Features

Research demonstrates strong correlations between economic indicators and power demand[20][4][21]:

### Economic Indicators

- **GDP Growth Rate**: Strong predictor of electricity demand[22][23]
- **Industrial Production Index**: Manufacturing activity correlation[24][25]
- **Consumer Price Index**: Economic activity indicator[24]
- **Unemployment Rate**: Affects residential and commercial demand[24][26]
- **Energy Prices**: Gas, oil prices affecting substitution effects[24]

### Demographic Features

- **Population Density**: Urban vs. rural consumption patterns[21][27]
- **Age Distribution**: Different age groups have different consumption patterns[21]
- **Income Levels**: Higher income correlates with higher consumption[21][26]
- **Education Levels**: Influences energy efficiency awareness[26]
- **Building Characteristics**: Age, type, size of buildings[26][28]

### Signal Decomposition Features

Advanced signal processing techniques significantly improve forecasting accuracy[29][30][31]:

### Decomposition Methods

- **Empirical Mode Decomposition (EMD)**: Separates intrinsic mode functions[30][32]
- **Variational Mode Decomposition (VMD)**: More robust than EMD[33][31]
- **Wavelet Decomposition**: Multi-resolution analysis[34][35][36]
- **Fourier Transform Features**: Frequency domain characteristics[37][38][39]

### Decomposition-Derived Features

- **Trend Components**: Long-term patterns[29]
- **Seasonal Components**: Regular periodic patterns[29]
- **Residual Components**: Irregular variations[29]
- **High/Low Frequency Decompositions**: Different time scales[40][31]

### Spatial and Geographic Features

For regional forecasting, spatial features provide significant improvements[41][42][43]:

### Spatial Correlation Features

- **Neighboring Region Loads**: Spatial autocorrelation[43]
- **Distance-Weighted Features**: Geographic proximity effects[41]
- **Land Use Classifications**: Residential, commercial, industrial zones[41][42]
- **Remote Sensing Data**: High-resolution satellite imagery features[41][42]

### Infrastructure Features

- **Grid Topology**: Network connectivity patterns[44]
- **Transmission Constraints**: Capacity limitations[45]
- **Distributed Generation**: Solar PV, wind penetration[46]

### Advanced Feature Interactions

Research shows interaction features can improve model performance significantly[47][48]:

### Feature Interactions

- **Temperature × Day Type**: Weekend vs. weekday temperature effects[19]
- **Weather × Economic**: Economic activity moderated by weather[2]
- **Time × Geographic**: Regional time zone effects[41]
- **Load × Weather Lag**: Thermal inertia effects[9]

### Dynamic Feature Engineering

- **Adaptive Features**: Features that change based on recent patterns[49]
- **Context-Aware Features**: Features dependent on current conditions[50]
- **Regime-Dependent Features**: Different features for different market conditions[51]

### Data-Driven Feature Discovery

Recent advances use AI for automatic feature discovery[52][51]:

### Automated Feature Engineering

- **LLM-Enhanced Features**: Using language models for feature extraction[51][53]
- **Genetic Algorithm Features**: Evolutionary feature construction[54]
- **Deep Feature Learning**: Automatic feature extraction via neural networks[49][55]

### Model Recommendations

Based on the research, these models consistently perform well for power demand forecasting:

### Top Performing Models

1. **Hybrid CNN-LSTM**: Captures both spatial and temporal patterns[56][57][55]
2. **Transformer-based Models**: Best for long sequences[58][59]
3. **XGBoost/LightGBM**: Excellent for tabular data with feature engineering[60][61][54]
4. **Ensemble Methods**: Combining multiple models[61][62]
5. **BiLSTM**: Bidirectional temporal dependencies[63][55]

### Preprocessing Recommendations

- **Feature Selection**: Use correlation analysis, mutual information[64][5]
- **Dimensionality Reduction**: PCA for high-dimensional features[47]
- **Normalization**: Critical for neural networks[65]
- **Missing Value Handling**: Advanced imputation techniques[49]

### Implementation Priority

Start with these high-impact features:

1. **Cooling/Heating Degree Days** - Most cited weather enhancement
2. **Holiday Indicators** - Can improve accuracy by 10-80%
3. **Signal Decomposition** - Trend, seasonal, residual components
4. **Economic Indicators** - GDP, unemployment rate
5. **Advanced Lag Features** - Multi-scale temporal dependencies

This comprehensive feature engineering approach, combined with appropriate model selection, should significantly improve your power demand forecasting accuracy based on the latest research findings.